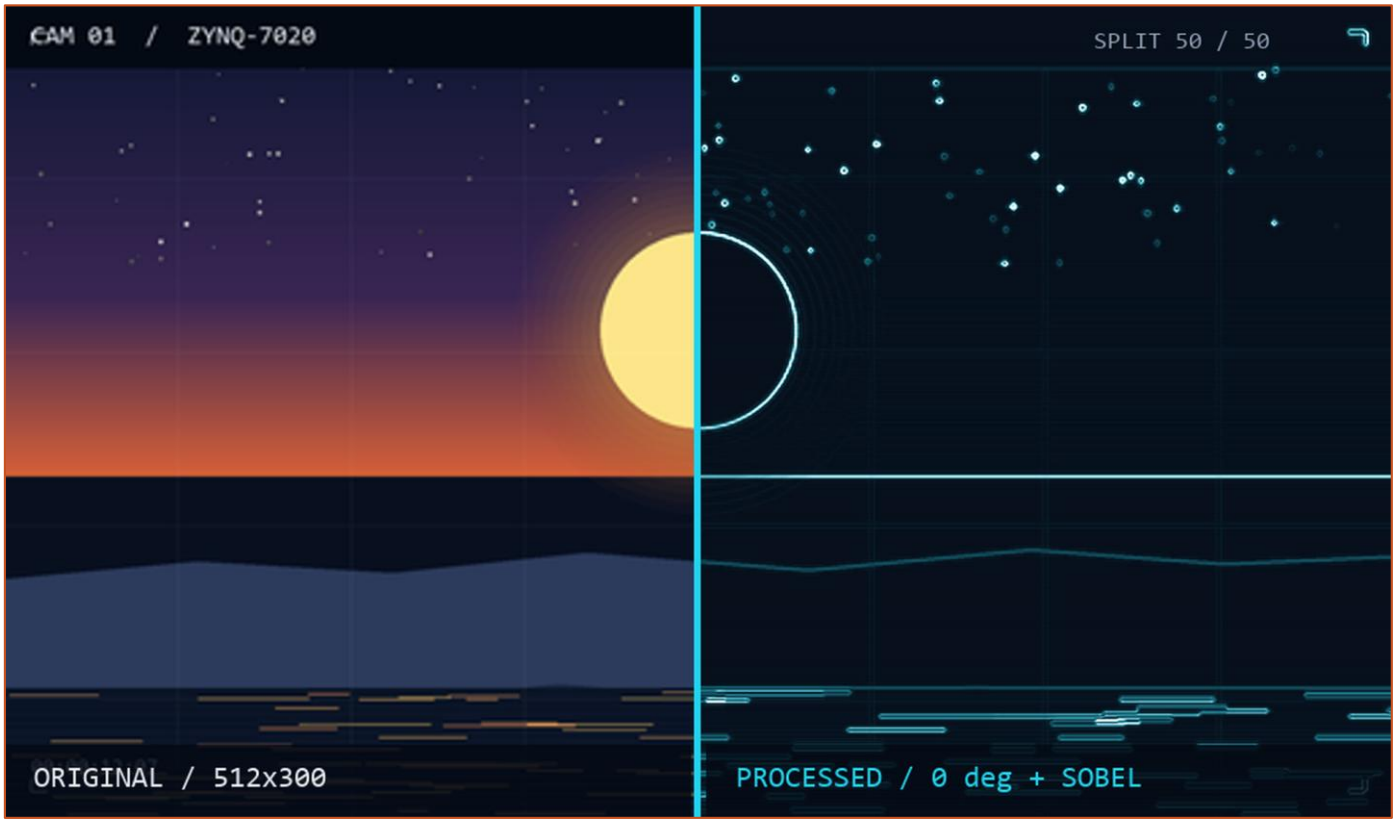
ZYNQ-7020 · A FULLY HARDWARE VIDEO NODE

# ONE FRAME.
# TWO TRUTHS.

Original and processed pixels leave one frame buffer in the same clock; the seam compares content, not screenshots. The node reports what it did on the wire at 1 Hz.

**SAME 512×300 FRAME / SOBEL / 0° / NO ROTATION**




*Rendered from the same test frame; two panes, one frame identity.*

## 01 SAME-FRAME A/B

Fixed 15-cycle chain; the seam is a pixel decision.

## 02 TELEMETRY

Frame-commit gates (bytes/rows/bad), aborts, latency, pane, gaps at 1 Hz.

## 03 FAULT INJECTION

Drop 1 in 100; aborts move; recovery returns.

**1.341 ns**
setup WNS · 0 failing endpoints

**0.025 ns**
hold WHS · tightest path

**70.00%**
BRAM · 98 of 140 tiles

**15 cycles**
fixed effect latency

**RGMII → MAC → UDP → DDR → GEOMETRY → 6-STAGE EFFECTS → A/B SEAM → HDMI**

512×300 RGB565 at 30 fps → 1024×600 at 59.5 Hz · xc7z020clg484-2 · 1 GbE UDP ingest · SD and test-pattern fallback

**29.95 FPS · PANE_BAD=0 · QSPI COLD BOOT**

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